

Pre-Generation LLM Failure Prediction Under Distribution Shift

Yael Moros-Daval¹, Álvaro David Gómez Antón¹, Kexin Jiang-Chen¹, Cèsar Ferri¹, José Hernández-Orallo^{1,2}, and Fernando Martínez-Plumed¹

¹ VRAIN, Universitat Politècnica de València, València, Spain

² Leverhulme Centre for the Future of Intelligence, University of Cambridge
ymordav@inf.upv.es, {adgomant,kjiache}@etsinf.upv.es,
{cferri,jorallo,fmartinez}@dsic.upv.es

Abstract. Reliable deployment of LLMs requires anticipating failures when the model is applied to tasks or prompts that differ from those seen during development. We study *pre-generation failure prediction*, where an auxiliary *assessor model* estimates the risk of an incorrect answer from the prompt alone. We evaluate it across four LLMs and six reasoning benchmarks using a leave-one-benchmark-out protocol. Because the benchmarks span disjoint task families (arithmetic, word puzzles, geographic recall, multiple-choice science, geometry, logic) with almost no shared vocabulary or format, this is a near-worst-case transfer test where chance is the natural null; what matters is which representations exceed chance under shift, not their absolute level. Dense embeddings are strongest in-domain but lose most of their advantage under shift, whereas interpretable cognitive-demand features degrade less sharply. The best out-of-domain results combine the two. Benchmark- and model-level analyses reveal substantial heterogeneity. Robust failure prediction therefore cannot rely on semantic representations alone: task-structural features add complementary, shift-resistant signal for interpretable reliability assessment.

Keywords: LLM reliability · Assessor Models · Distribution Shift · ADeLe

1 Introduction

Large language models (LLMs) are increasingly embedded in systems where their outputs influence real decisions, making confidently fluent but incorrect answers a significant reliability concern. A natural question is therefore: *will the model get this prompt wrong?* If failure can be anticipated before generation, the system can abstain, defer to a human, or route the request to a stronger model, avoiding the cost and risk of producing an unreliable answer. This setting falls within the classical *reject option* framework [3,4,8], which we instantiate through a *pre-generation assessor*: a lightweight auxiliary model that estimates failure risk from the prompt alone [14,26].

Failure prediction is a supervised classification problem whose performance can degrade substantially when the deployment distribution differs from the training distribution, a phenomenon known as *dataset shift* [21]. This challenge is particularly relevant for LLM reliability, since assessors are most valuable precisely when deployed on tasks that differ from those used during development. The key question is therefore not only whether failures can be predicted, but whether the signals used to predict them transfer across task families. In this work, we study pre-generation failure prediction under benchmark shift and investigate which prompt representations remain informative when the assessor is applied to previously unseen tasks.

To study this question, we evaluate failure prediction across four LLMs and six reasoning benchmarks using a leave-one-benchmark-out protocol. In each experiment, the assessor is trained on five task families and evaluated on a previously unseen sixth, creating a controlled benchmark-shift setting that tests whether failure signals transfer beyond the tasks on which they were learned. Our contributions are:

1. **Pre-generation failure prediction under benchmark shift.** We study LLM failure prediction as a transfer problem, evaluating whether a pre-generation assessor trained on some task families can anticipate failures on previously unseen ones. To this end, we introduce a leave-one-benchmark-out protocol spanning four LLMs and six reasoning benchmarks.
2. **Characterising the shift.** We choose six maximally disjoint task families, so leave-one-benchmark-out induces a near-worst-case shift: train and test prompts share almost no vocabulary, format, or solution schema, combining near-total covariate shift with large prevalence differences. The transfer we measure is thus a conservative lower bound on milder, within-family deployment drift.
3. **The limits of embedding-based transfer.** Dense semantic embeddings achieve the strongest in-domain discrimination, but much of this advantage disappears under benchmark shift. Strong in-domain failure prediction therefore does not necessarily translate into transferable failure prediction.
4. **Cognitive-demand features are less sensitive to benchmark shift.** While cognitive-demand features achieve only modest out-of-domain discrimination on their own, their performance degrades less sharply than that of embeddings. When combined with embeddings, they consistently improve out-of-domain results, suggesting that they capture aspects of failure risk that are less benchmark-specific than purely semantic representations.
5. **Benchmark- and model-level analysis of failure predictability.** We find that transferability varies substantially across both task families and LLMs. Some benchmarks remain near chance regardless of representation, while others benefit strongly from cognitive-demand information; similarly, some models exhibit much more predictable failure patterns than others.

2 Related work

2.1 Selective prediction and failure assessment for LLMs

Selective prediction studies systems that may abstain when their predictions are likely to be unreliable, trading coverage for accuracy through a reject option [3,4]. In modern neural models, abstention is commonly implemented either as a post-hoc policy over a fixed predictor [8] or as an explicit selection mechanism trained jointly with the predictor [9]. Related work has also explored calibration [13], uncertainty estimation [7,18], and conformal prediction [27,1] as mechanisms for identifying predictions on which a model should abstain.

For LLMs, reliability assessment is complicated by free-form generation, semantic equivalence, and reasoning variability. Existing approaches estimate correctness either after generation, using confidence scores, self-evaluation, agreement across samples, or semantic uncertainty [16,20,17,5]; or before generation through assessor models and related predictive approaches [14,26,25,19]. Such methods support early abstention, escalation to stronger models, or routing in multi-model systems [22,15].

2.2 Dataset shift and domain generalisation

A central challenge in machine learning is that models are often deployed on data that differ from those seen during training, a phenomenon broadly referred to as *dataset shift* [21]. Under such shifts, predictors may exploit patterns that are highly informative in the source domain but fail to generalise to new environments. This problem has motivated extensive work on domain adaptation, domain generalisation, and robust learning, all of which seek representations that remain predictive when the data distribution changes [32,30].

Dataset shift is particularly relevant for LLM reliability assessment. An assessor trained on one collection of tasks may encounter prompts with different linguistic characteristics, reasoning requirements, and error rates when deployed on another. Consequently, strong in-domain performance does not necessarily imply reliable failure prediction on unseen tasks. Recent work has highlighted the sensitivity of uncertainty estimates, confidence scores, and selective prediction mechanisms to distribution shift [6,31], motivating evaluations that explicitly measure transfer beyond the training distribution. From this perspective, an assessor built on dense embeddings is prone to rely on *shortcut* features—surface regularities such as digit strings or proper nouns that correlate with failure within a task family but carry no signal once the family changes—mirroring the shortcut-learning and spurious-correlation phenomena documented in supervised models [10,2].

Our work studies this problem through benchmark shift: assessors are trained on a set of reasoning task families and evaluated on a previously unseen one. Rather than proposing a new adaptation method, we investigate which prompt representations are most resistant to this shift and whether interpretable task-level information degrades differently from semantic embeddings when transferred across domains.

3 Methodology

3.1 LLM failure as a classification problem

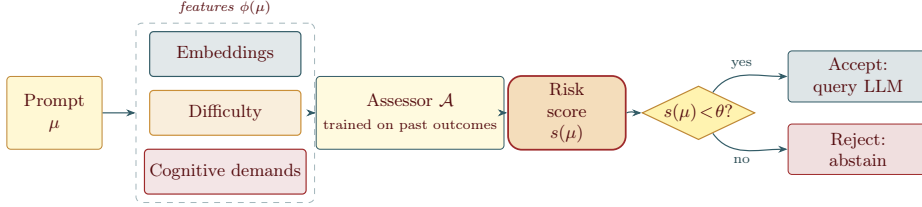


Fig. 1: Pre-generation failure assessment. Three descriptions of the prompt (embeddings, intrinsic difficulty, and ADeLe cognitive demands) feed a small trained assessor that outputs a risk score $s(\mu)$ before the LLM is run. Thresholding this score gives a reject rule: prompts predicted to be safe are sent to the LLM, while high-risk prompts are rejected, deferred, or routed elsewhere.

Let $\mathcal{M}$ denote an LLM and let $\mu_{i=1}^N$ be a benchmark of prompts. For each prompt, an external automatic procedure labels the model’s response as *correct* (C), *incorrect* (I), or *avoidance* (A), where the model explicitly declines to answer. We regard C and A as *safe* outcomes, since both a correct answer and an honest refusal are acceptable, whereas I is considered *unsafe*. A *pre-generation assessor* is a model $\mathcal{A}$ that predicts the risk associated with a prompt before the LLM is queried [14]. Specifically, it maps prompt features $\phi(\mu)$ to a risk score $s(\mu) \in [0, 1]$ without accessing the response that $\mathcal{M}$ would generate. Given a threshold θ , the resulting abstention policy allows $\mathcal{M}$ to answer when $s(\mu) < \theta$ and abstains otherwise. Prompts rejected by the policy are assigned the label R. Figure 1 summarises this pipeline.

3.2 Prompt feature families

The assessor input, $\phi(\mu)$, is a feature-based representation of the prompt. We consider three feature families and evaluate seven feature views: each family individually, each pairwise combination, and the combination of all three families.

Embeddings (Emb). A dense vector representation of the prompt produced by a strong text encoder [23]. This representation captures both semantic content and surface-level characteristics, such as vocabulary and phrasing. Embeddings provide a rich and highly informative description of the prompt, but they may also encode benchmark-specific patterns that do not necessarily transfer across tasks, making them potentially sensitive to covariate shift.

Intrinsic difficulty (Diff). A scalar feature intended to capture the intrinsic difficulty of a prompt independently of any particular LLM. The difficulty proxy is task-specific and derived from properties of the problem itself, such as the number of carry operations (addition), word length (anagram), inverse city popularity (locality), estimated human difficulty (science), or problem size (e.g., grid or polygon size in structured tasks). This representation is transparent and readily interpretable, but necessarily coarse: a single scalar can indicate that a prompt is difficult without explaining the source of that difficulty. Moreover, because the proxy is defined separately for each benchmark, its interpretation may not transfer consistently across tasks.

Cognitive demands (ADeLe). ADeLe represents a prompt through an interpretable profile of the abilities required to solve it [33]. Rather than describing the content or surface form of a prompt, it characterises the extent to which different cognitive abilities and knowledge types are required for successful completion.

Each prompt is assigned a score from 0 to 5 on 18 dimensions using explicit rubric-based criteria (Table 1). These dimensions comprise 11 cognitive dimensions grouped into seven capability families, 5 knowledge dimensions, and 2 additional dimensions capturing *atypicality* (how unusual an instance is) and *volume* (the amount of information to process). The scores are automatically assigned by an LLM annotator using the ADeLe rubrics, yielding an 18-dimensional feature vector for each prompt.

ADeLe has two properties that make it attractive for transfer. First, it is *interpretable*: the representation identifies which abilities contribute to the difficulty of a prompt (e.g., high logical reasoning and high atypicality), making risk predictions auditable along meaningful dimensions. Second, it is *task-structural and model-agnostic*: it describes the abilities required to solve a task rather than its surface form. Consequently, the meaning of a given demand level is intended to remain stable across task families. Whereas two prompts with similar difficulty may require very different skills, ADeLe explicitly captures this structure, which we hypothesise is more likely to transfer across tasks.

3.3 Evaluation measures and protocols

Consider a labelled evaluation set of N prompts with outcomes $y(\mu) \in \{\mathbf{C}, \mathbf{l}, \mathbf{A}\}$. Let $\mathcal{S} = \{\mu : y(\mu) \in \{\mathbf{C}, \mathbf{A}\}\}$ denote the set of *safe* prompts and $\mathcal{U} = \{\mu : y(\mu) = \mathbf{l}\}$ the set of *unsafe* prompts. The true error prevalence is therefore $p = |\mathcal{U}|/N$.

The assessor assigns each prompt a risk score $s(\mu) \in [0, 1]$, where larger values indicate higher estimated failure risk. Given a threshold θ , the policy answers when $s(\mu) < \theta$ and abstains otherwise. This induces a rejection rate $r(\theta) = \frac{1}{N} \sum_{\mu} \mathbf{1}[s(\mu) \geq \theta]$, and coverage $\text{Cov} = 1 - r$.

We evaluate the assessor as a classifier that distinguishes safe from unsafe prompts using Eqs. (1)–(3).

Table 1: The 18 ADeLe demand dimensions used as features, with their short codes and rubric descriptions [33]. Each prompt is scored 0–5 on every dimension by an LLM annotator following explicit per-level rubrics. In the cognitive block, shared code prefixes mark the seven capability families (**CE**: comprehension/expression; **MC**: metacognition/critical thinking; **QL**: quantitative/logical reasoning).

Code	Dimension	Demand description
<i>Cognitive demands (11)</i>		
AS	Attention & Scan	Focus on or locate specific elements within a stream of information or environment while solving the task.
CEc	Verbal Comprehension	Understand text, stories, or the semantic content of other representations of ideas across formats and modalities.
CEe	Verbal Expression	Generate and articulate ideas, stories, or semantic content across formats and modalities.
CL	Conceptualisation, Learning & Abstraction	Build new concepts, reason inductively and by analogy, map relations between domains, and abstract from concrete examples.
MCr	Identifying Relevant Information	Recognise which information does or does not help solve the task, and how that recognition unfolds while working toward a solution.
MCt	Critical Thinking Processes	Monitor and regulate multiple thought processes, from simple recall to high-level critical thinking.
MCu	Calibrating Knowns & Unknowns	Recognise the boundaries of one’s own knowledge: what one knows, does not know, or is uncertain about.
MS	Mind Modelling & Social Cognition	Model other agents’ beliefs, desires, intentions, and emotions, and how they interact to shape behaviour.
QLl	Logical Reasoning	Match and apply rules, procedures, or algorithms to premises to derive conclusions and make decisions.
QLq	Quantitative Reasoning	Work with and reason about quantities, numbers, and numerical relationships.
SNs	Spatio-physical Reasoning	Understand spatial relationships between objects and predict physical interactions.
<i>Knowledge demands (5)</i>		
KNa	Applied Sciences	Knowledge or conceptual understanding in applied sciences (medicine, law, education, business, engineering, ...).
KNe	Customary Everyday Knowledge	Information most people in a society acquire through daily life, social interaction, and media.
KNf	Formal Sciences	Knowledge or conceptual understanding in formal sciences (mathematics, logic, computer science, statistics).
KNn	Natural Sciences	Knowledge or conceptual understanding in natural sciences (physics, chemistry, biology, astronomy, ...).
KNs	Social Sciences & Humanities	Knowledge or conceptual understanding in social sciences and humanities (history, psychology, literature, philosophy, ...).
<i>Extraneous demands (2)</i>		
AT	Atypicality	How uncommon the task is: how unlikely it is that the instance has appeared in sources such as the internet, textbooks, or tests.
VO	Volume	Proportional to the logarithm of the time a fully competent human needs to read and complete the task under ideal conditions.

$$\text{S-Acc}(\theta) = \frac{1}{|S|} \sum_{\mu \in S} \mathbf{1}[s(\mu) < \theta], \quad \text{U-Rej}(\theta) = \frac{1}{|\mathcal{U}|} \sum_{\mu \in \mathcal{U}} \mathbf{1}[s(\mu) \geq \theta], \quad (1)$$

$$\text{FOC-AUC} = \Pr_{u \in \mathcal{U}, v \in S} [s(u) > s(v)], \quad (2)$$

$$\text{Ret} = \frac{\text{FOC-AUC}^{\text{OOD}} - \frac{1}{2}}{\text{FOC-AUC}^{\text{ID}} - \frac{1}{2}}. \quad (3)$$

Operating-point rates. At a fixed threshold θ , S-Acc is the fraction of safe prompts that remain answerable under the policy, while U-Rej is the fraction of unsafe prompts on which the policy abstains (see eqs. (1)). Equivalently, these are the recalls of the safe and unsafe classes, respectively. Both metrics lie in $[0, 1]$, and increasing one typically decreases the other: a stricter threshold raises U-Rej by catching more errors, but lowers S-Acc by refusing more safe responses.

Threshold-free discrimination. Varying θ from accept-all to reject-all traces the *filter operating characteristic* (FOC), which plots S-Acc against U-Rej. Its area, FOC-AUC (see eq. (2)), summarises discrimination independently of any particular operating point. By the Mann–Whitney identity, FOC-AUC equals the probability that a randomly chosen unsafe prompt receives a higher risk score than a randomly chosen safe prompt, with ties counted as $\frac{1}{2}$. Equivalently, it is the AUROC of the unsafe-versus-safe ranking. A value of 1 indicates perfect separation, while $\frac{1}{2}$ corresponds to chance performance. We use FOC-AUC as our primary classification metric. In practice, θ is an application-specific design choice that reflects the desired trade-off between answering more prompts and avoiding unsafe responses.

Transfer. Retention measures how much discrimination survives a distribution shift. It compares out-of-domain (OOD) and in-domain (ID) performance after subtracting the chance baseline of $\frac{1}{2}$ (see eq. (3)). Thus, $\text{Ret} = 1$ (100%) indicates no loss in discrimination, $\text{Ret} = 0$ corresponds to a collapse to chance performance, and $\text{Ret} < 0$ indicates worse-than-chance transfer. Subtracting the chance floor makes retention comparable across tasks with different in-domain difficulty levels, which a raw ratio of FOC-AUCs would not.

Protocols. We use two regimes. *In-domain* (ID): train and test on disjoint splits of the *same* benchmark, so train and test are identically distributed. *Cross-domain* (OOD): *leave-one-benchmark-out*, train on five benchmarks and test on the held-out sixth, so test prompts come from an unseen task family. The OOD protocol is our controlled dataset shift. Unless stated otherwise, FOC-AUC is reported as the average across the four LLMs, both per held-out benchmark and pooled.

4 Experimental setup

Models and benchmarks. We use four widely deployed LLMs, two open-weight and two proprietary: LLaMA-3.3-70B [12], Qwen-2.5-72B [29], GPT-5.2 [24], and Gemini-3-Pro [11], queried at temperature 0. We evaluate on six reasoning benchmarks, each with a per-prompt intrinsic difficulty proxy (Table 2): four from a reliability suite (arithmetic, word puzzles, geographic knowledge, multiple-choice science) [34] and two families of programmatically generated structured-reasoning problems (geometry, logic puzzles) [28]. Answers are labelled C/I/A by task-specific automatic procedures verified against human checks. To make difficulty comparable across the two benchmark sources in the OOD setting,

difficulty is mapped onto a common scale by a monotone calibration fitted on a reference model’s success rates.

Table 2: The six reasoning benchmarks, their task types, and the intrinsic prompt difficulty proxy $\delta(\mu)$. Difficulty is computed from the prompt alone and used as the Diff feature. Benchmarks 1–4 are from ReliabilityBench [34]; benchmarks 5–6 are structured-reasoning tasks from OMEGA [28].

Benchmark	Task (what the prompt asks)	Difficulty proxy $\delta(\mu)$
Addition [34]	Add two integers of up to 100 digits.	δ = number of carry operations in the sum $a+b$.
Anagram [34]	Unscramble a letter string into an English word w .	$\delta = w $, the word length (permutations grow as $ w !$).
Locality [34]	Name the most populous city within a fixed radius of a given city.	δ = inverse popularity (population rank) of the answer city; rarer cities are harder.
Science [34]	Answer a multiple-choice science question (school to graduate level).	δ = the item’s annotated human-difficulty level.
Geometry [28]	Counting/colouring problems on a regular polygon.	$\delta = n$, the polygon’s number of sides ($n(n-3)/2$ diagonals).
Logic Puzzles [28]	Grid paths, “zebra”-style constraint puzzles, chip placement.	δ = number of constrained entities (grid size).

5 Results

5.1 Characterising the shift

Before evaluating transfer, we first characterise the distribution shifts induced by the leave-one-benchmark-out protocol. Since the assessor’s target is the unsafe outcome $\mathbf{l}$, the relevant class prior is the prevalence of incorrect responses, $p = P(\mathbf{l})$.

Figure 2 reports this prevalence for every (model, benchmark) pair, corresponding to the LLM’s baseline error rate before any abstention ($r = 0$). The resulting variation is substantial: error rates range from 0.001 (**Gemini-3-Pro** on **Anagram**) to 0.86 (**Qwen-2.5-72B** on **Anagram**), with a mean of 0.44. The variation is large both across models and across tasks. Within a single benchmark, error prevalence can span nearly the entire unit interval (e.g., 0.001–0.86 on **Anagram**), while within a single model it can vary dramatically across benchmarks (e.g., **LLaMA** ranges from 0.19 on **Science** to 0.79 on **Logic Puzzles**).

The leave-one-benchmark-out protocol induces substantial distribution shift. Because the assessor for a held-out benchmark is trained on the other five, the prior it encounters during training is approximately their average, whereas the target prior is that of the held-out task. Treating benchmarks as equal-sized, the resulting prior-probability shift $|\Delta p| = |p_{\text{test}} - \bar{p}_{\text{train}}|$ averages 0.15 across the 24 (model, benchmark) cells and reaches 0.48 (Table 3).

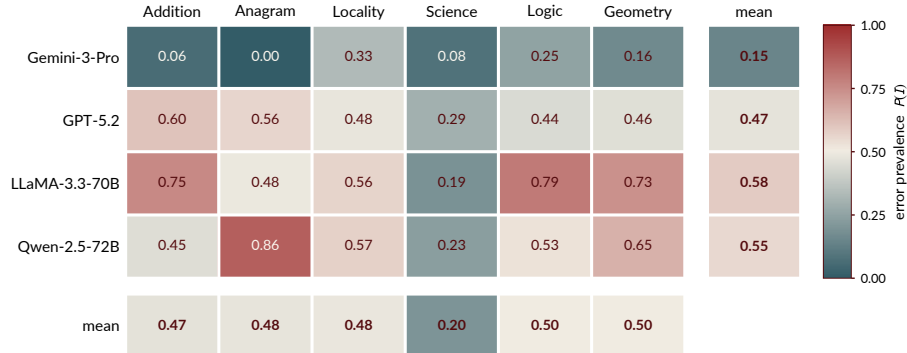


Fig. 2: Base error prevalence $p = P(I)$ per model and benchmark (the base error rate the assessor must anticipate). It spans 0.001–0.86, so leaving out one benchmark moves the class prior far from the training mix: this is severe *prior-probability (label) shift*.

Table 3: Per-benchmark prevalence and the prior-probability shift induced by leave-one-benchmark-out. $\bar{p}$: error prevalence pooled over the four LLMs. $|\Delta p|$: mean over LLMs of $|p_{\text{test}} - \bar{p}_{\text{train}}|$ when that benchmark is held out.

	Addition	Anagram	Locality	Science	Logic	Geometry	mean
$\bar{p}$ (prevalence)	0.47	0.48	0.48	0.20	0.50	0.50	0.44
$ \Delta p $ (shift)	0.15	0.19	0.07	0.29	0.11	0.09	0.15

The largest shifts occur for **Science**, whose low error rates ($\bar{p} \approx 0.20$) differ substantially from those of the remaining benchmarks ($\bar{p} \approx 0.47\text{--}0.50$), yielding an average shift of 0.29. This label shift is accompanied by strong covariate shift: prompt distributions differ sharply across task families, ranging from digit strings and city names to polygons and logical puzzles. The leave-one-benchmark-out protocol, therefore, constitutes a genuinely challenging dataset-shift setting rather than a mild train–test shift. Because the six families share essentially no surface vocabulary, input format, or solution procedure, the covariate supports of the training mix and the held-out task are close to disjoint. Under such near-total covariate shift, chance-level transfer is the appropriate null expectation for any prompt representation, and the substantive question is not whether out-of-domain discrimination is high in absolute terms (few representations could be) but which representation exceeds chance at all, and why.

5.2 Classification under shift: which features transfer?

We now ask which prompt feature lets the assessor *classify* unsafe prompts across this shift, using the error-risk assessor $\mathcal{A}_{\text{ER}}$. Table 4 gives, per feature view, in-domain and cross-domain FOC-AUC, the drop, and retention; Table 5

breaks the cross-domain numbers down by held-out benchmark; Fig. 3 visualises the trade-off.

Because the leave-one-benchmark-out setting places train and test on near-disjoint task supports (Section 5.1), an out-of-domain FOC-AUC near 0.5 is the expected null rather than a sign of method failure. We therefore read Table 4 primarily through *retention*, i.e., the fraction of above-chance discrimination that survives the shift, and through the *ordering* of feature views, rather than through absolute OOD levels.

Table 4: Average FOC-AUC (over four LLMs) for the error-risk assessor $\mathcal{A}_{\text{ER}}$ by feature view. **ID**: in-domain; **OOD**: leave-one-benchmark-out; **Δ** : drop; **Ret.**: above-chance discrimination retained. 0.5 is random. Best per block in bold.

Feature view	ID $\uparrow$	OOD $\uparrow$	Δ $\downarrow$	Ret. $\uparrow$
<i>Interpretable only (describe the task, not its words)</i>				
Difficulty (Diff)	0.613	0.495	0.118	−4.5%
Cognitive demands (ADeLe)	0.662	0.536	0.126	22.3%
Diff + ADeLe	0.696	0.522	0.174	11.0%
<i>Using embeddings (describe meaning and surface form)</i>				
Embeddings (Emb)	0.738	0.535	0.203	14.7%
Emb + Diff	0.746	0.528	0.217	11.5%
Emb + ADeLe	0.759	0.567	0.192	25.9%
Emb + Diff + ADeLe	0.769	0.569	0.201	25.5%

Embeddings dominate in-domain, but their advantage shrinks under shift. Embedding-based views achieve the strongest in-domain performance, reaching FOC-AUC values around 0.75 compared with 0.61–0.70 for interpretable-only views (Table 4). Under leave-one-benchmark-out evaluation, however, this advantage largely disappears. While embedding-based models attain the highest in-domain discrimination, their performance drops substantially under shift, yielding OOD scores comparable to those of cognitive-demand features. For example, embeddings alone achieve 0.535 OOD FOC-AUC, essentially matching ADeLe (0.536) despite a much larger in-domain advantage. This pattern suggests that embeddings capture highly predictive information that is benchmark-dependent, whereas cognitive-demand features encode weaker but more stable signals that transfer across tasks.

Cognitive-demand features provide complementary signal under shift. The leave-one-benchmark-out results show that failure prediction remains challenging under benchmark shift. Even the strongest feature views achieve only modest OOD discrimination (FOC-AUC $\approx$ 0.57), and several benchmark-feature combinations remain close to chance. Within this difficult setting, however, ADeLe features

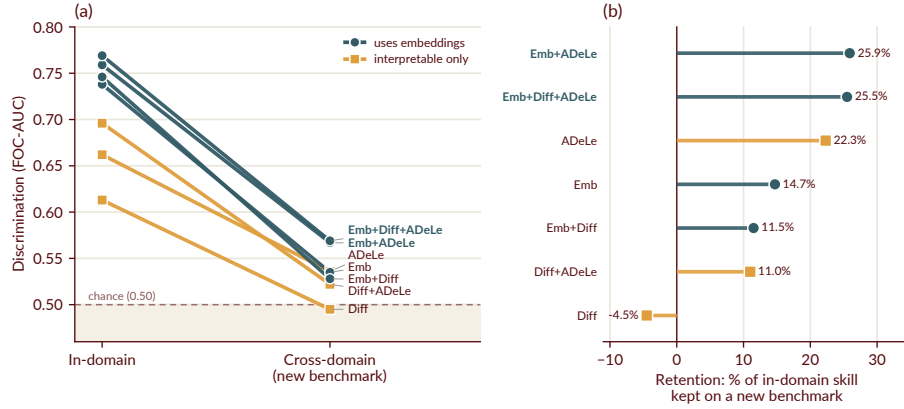


Fig. 3: Classification under shift, error-risk assessor. **(a)** Each line runs from a feature view’s in-domain to its cross-domain FOC-AUC: embedding views (teal) start highest and fall furthest; interpretable-only views (gold) start lower and fall less. **(b)** Above-chance discrimination retained under shift: adding ADeLe raises retention wherever applied; Emb+ADeLe keeps the most.

provide a consistent benefit. Adding cognitive-demand information improves embedding-based models, increasing OOD FOC-AUC from 0.535 to 0.567 for Emb→Emb+ADeLe and from 0.528 to 0.569 for Emb+Diff→Emb+Diff+ADeLe (Table 4). Among interpretable feature views, ADeLe alone achieves the strongest OOD performance, matching or slightly exceeding embeddings despite relying only on a compact, human-interpretable task description. The strongest transfer results overall are obtained by the ADeLe-bearing embedding models, Emb+ADeLe and Emb+Diff+ADeLe. Taken together, these findings suggest that cognitive-demand features capture aspects of failure risk that complement the information encoded by semantic embeddings.

Transfer behaviour varies substantially across benchmarks. The benchmark-level analysis (Table 5) reveals considerable heterogeneity in transfer performance. No feature view is best across all held-out benchmarks, and some tasks remain difficult regardless of representation. For example, performance on Locality is close to chance for all feature views, suggesting limited transferability of failure patterns from other domains. In contrast, other benchmarks show substantial differences between representations. On Anagram, embeddings alone perform below chance (0.467), whereas ADeLe-based models exceed 0.60. Geometry and Logic achieve their strongest results with embedding–ADeLe combinations. These patterns indicate that benchmark shift is not a single phenomenon: different task families preserve different aspects of the failure signal, and no representation is uniformly robust across all domains.

Table 5: Cross-domain (leave-one-benchmark-out) FOC-AUC of the error-risk assessor $\mathcal{A}_{\text{ER}}$ per held-out benchmark and feature view (average over LLMs). Best per column in bold. Embedding-bearing views are strongest in-domain (Table 4) yet here hover near chance on several tasks; demand-bearing views transfer more evenly. The final column reports the average over held-out benchmarks.

Feature view	Addition	Anagram	Locality	Science	Logic	Geometry	avg.
Diff	0.483	0.396	0.504	0.488	0.551	0.547	0.495
ADeLe	0.507	0.604	0.504	0.575	0.530	0.513	0.539
Diff+ADeLe	0.458	0.512	0.507	0.588	0.531	0.531	0.521
Emb	0.523	0.467	0.501	0.543	0.579	0.600	0.535
Emb+Diff	0.551	0.429	0.485	0.533	0.595	0.575	0.528
Emb+Diff+ADeLe	0.540	0.611	0.483	0.596	0.608	0.585	0.570
Emb+ADeLe	0.512	0.612	0.499	0.585	0.602	0.605	0.569

Table 6: Average FOC-AUC for the error-risk assessor $\mathcal{A}_{\text{ER}}$ per LLM and distribution.

Model	ID $\uparrow$	OOD $\uparrow$
gemini-3-pro-preview	0.680	0.511
GPT5.2	0.722	0.557
llama3v3chat-70B	0.714	0.589
Qwen2.5_72B	0.729	0.484

Failure predictability depends strongly on the target model. Variation across LLMs is larger than variation across feature views, as we see in Table 6. While average OOD discrimination ranges from 0.535 to 0.570 across feature representations, it ranges from 0.484 to 0.589 across models. The assessor is therefore substantially more effective for some LLMs than others. In particular, **Llama3v3-Chat-70B** achieves the strongest transfer performance (0.589 OOD FOC-AUC), whereas **Qwen2.5-72B** remains close to chance overall (0.484). This suggests that failure behaviour is not equally predictable across models: some LLMs exhibit error patterns that generalise across benchmarks, while others appear substantially harder to anticipate from prompt-level features alone.

Benchmark transfer is model-specific. The benchmark-level results reveal substantial interactions between task family and model (Table 7). **Llama3v3Chat-70B** achieves strong discrimination on Logic (0.681) and Geometry (0.663), whereas Gemini performs near chance on Anagram (0.222) despite competitive results on Science (0.615). **Qwen2.5-72B** remains close to chance on most held-out benchmarks. These patterns indicate that transferability is not solely a property of the benchmark or the feature representation. Instead, the structure of model failures appears to differ across LLMs, with some models exhibiting more stable and predictable error patterns than others.

Table 7: Average OOD FOC-AUC for the error-risk assessor $\mathcal{A}_{\text{ER}}$ per LLM and held-out benchmark.

Model	Addition	Anagram	Locality	Science	Logic	Geometry
Gemini-3-Pro	0.462	0.222	0.501	0.615	0.579	0.519
GPT-5.2	0.567	0.582	0.496	0.556	0.570	0.569
LLaMA-3.3-70B	0.549	0.548	0.504	0.588	0.681	0.663
Qwen-2.5-72B	0.464	0.516	0.489	0.474	0.453	0.510

6 Discussion

Representation governs transferability. For pre-generation LLM failure prediction, the central challenge is not achieving high in-domain discrimination but maintaining it under benchmark shift. Embedding-based representations are strongest in-domain, yet much of this advantage disappears in leave-one-benchmark-out evaluation. Cognitive-demand features transfer more robustly, achieving OOD performance comparable to embeddings despite relying on a compact, interpretable description of the task. However, this robustness should not be overstated: ADeLe alone reaches only modest discrimination (FOC-AUC ≈ 0.54), and even the strongest feature combinations remain around 0.57 OOD FOC-AUC. Failure prediction therefore remains difficult under benchmark shift. The main contribution of cognitive-demand features is not to solve the transfer problem, but to provide complementary information that consistently improves embedding-based assessors. The strongest OOD results are obtained by Emb+ADeLe and Emb+Diff+ADeLe, suggesting that robust failure prediction benefits from combining semantic and task-structural information rather than relying on either alone. Because our leave-one-benchmark-out suite is deliberately adversarial, these OOD figures should be read as a lower bound: the differential robustness of task-structural features, rather than their absolute level, is the property most likely to carry over to the milder shifts seen in practice.

Practical guidance. Our results suggest different feature choices for different deployment scenarios. When the deployment distribution closely matches the training distribution and labelled data are available, embedding-rich assessors provide the strongest per-instance predictions. Under task drift, however, their advantage shrinks considerably. In such settings, augmenting embeddings with cognitive-demand features yields the most reliable transfer performance, while also providing an interpretable description of why a prompt may be risky. At the same time, practitioners should calibrate expectations: even the strongest feature views achieve only modest OOD discrimination (FOC-AUC ≈ 0.57), indicating that benchmark-shifted failure prediction remains a difficult problem. Rather than eliminating uncertainty, cognitive-demand features provide a small but consistent improvement in robustness across unseen tasks.

Limitations. Several limitations should be considered. Difficulty proxies are benchmark-specific and coarse, particularly for the structured tasks where difficulty is represented by a small number of discrete levels. Our cross-domain protocol leaves out one benchmark at a time from a suite of deliberately disjoint task families that share almost no vocabulary, format, or solution schema. This makes leave-one-benchmark-out a particularly stringent transfer setting: the assessor is trained and tested on different task families with largely non-overlapping prompt distributions, rather than the more gradual within-task drift commonly encountered in deployment. The transfer we report should therefore be read as a conservative lower bound: under more realistic, related-domain shift we would expect more in-domain signal (including part of the embedding advantage) to survive. The flip side is that this suite cannot characterise that milder regime.

Transfer performance also varies across benchmarks and models. Some benchmark-model combinations achieve useful discrimination, whereas others remain near chance, suggesting that the predictability of LLM failures is itself model-dependent. Models with more stable relationships between prompt characteristics and failure appear to admit more transferable assessors, although the reasons remain an open question. All results rely on LLM-generated cognitive-demand annotations; future work should explore alternative annotation procedures and broader benchmark collections.

7 Conclusion

We studied pre-generation LLM failure prediction under benchmark shift using task difficulty, cognitive-demand profiles, and semantic embeddings. A leave-one-benchmark-out evaluation revealed that transfer remains challenging: feature views that achieve strong in-domain discrimination often degrade substantially when applied to unseen task families. In particular, embeddings provide the strongest in-domain performance but lose much of their advantage under shift.

Our results show that cognitive-demand features offer a complementary and interpretable source of information about failure risk. Although ADeLe alone achieves only modest OOD discrimination, it consistently improves embedding-based assessors and contributes to the strongest transfer performance observed in our experiments. The benchmark-level analysis further reveals that transferability is highly heterogeneous, varying across both task families and target models.

Overall, our findings suggest that robust failure prediction cannot rely solely on semantic representations of prompts. Information about what a task requires provides additional signal that remains useful under distribution shift. While current performance remains far from sufficient for reliable deployment, combining semantic and task-structural representations appears to be a promising direction for more transferable and interpretable LLM reliability assessment.

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